CAPSTONE 1: SYSTEM ANALYSIS AND DESIGN REPORT

ONLINE SHOPPING SYSTEM

User Registration and Purchasing Subsystems

Course Title	Capstone 1 – System Analysis and Design	**Project Team**	Galautdinova Ruzana, Rugacheva Arina, Kan Daniil
Project Title	Online Shopping System	**Focus Area**	User Registration & Purchasing Subsystems

1. PROJECT OVERVIEW

In this project, our team designed and analyzed a comprehensive Online Shopping System aimed at simplifying retail purchasing for customers while providing backend tools for system administrators. We focused on building a secure, user-friendly platform that handles registration, product browsing, shopping cart updates, checkout processing, third-party payment integration, and order tracking.

System Workflow: Our designed system allows customers to create an account, authenticate, and search through items organized by categories. When a customer adds items to their shopping cart and proceeds to checkout, the system collects shipping information and interacts with an external Payment Service to verify payment. Once confirmed, the system generates an order, decrements the available product stock, and provides tracking status updates to the customer.

2. PROJECT SCOPE & BOUNDARY DEFINITIONS

IN-SCOPE SYSTEM CAPABILITIES:

- Customer account registration, password hashing, and login authentication.
- Product catalog browsing, category filtering, and keyword search.
- Interactive shopping cart management (add, edit item quantities, remove).
- Checkout workflow capturing recipient shipping addresses.
- Payment gateway interface with third-party service confirmation.
- Automatic real-time stock quantity updates after successful order placement.
- Customer order status tracking and purchase history logs.
- Admin management tools for products, inventory stock, and order status.

OUT-OF-SCOPE FUNCTIONS:

- Multi-vendor seller marketplace management (single storefront only).
- Native mobile app development for iOS and Android.
- Complex warehouse logistics and physical delivery route optimization.
- Customer product rating and written review moderation tools.
- Loyalty reward points and dynamic promo code engines.

3. SYSTEM ACTORS & ROLE RESPONSIBILITIES

Actor Name	Actor Type	Responsibilities & Interactions
Customer	Primary End User	Registers profile, logs in, searches catalog, manages cart items, enters shipping information, completes payments, and tracks placed orders.
Administrator	Internal System Manager	Manages product listings (CRUD), updates inventory stock levels directly, reviews orders, updates delivery fulfillment states, and manages catalog categories.
Payment Service	External System Gateway	Receives encrypted payment data, verifies cardholder funds, processes billing authorizations, and returns transaction status responses (`SUCCESS` / `FAILED`).

2.4 FUNCTIONAL REQUIREMENTS (FR)

Req ID	Module / Subsystem	Requirement Specification Description
FR-01	User Management	System shall allow visitors to register an account using a unique email address and password.
FR-02	User Management	System shall authenticate user credentials and issue encrypted session tokens.
FR-03	Catalog Search	System shall allow users to search products via keywords and filter listings by category.
FR-04	Catalog Display	System shall render detailed product views including SKU, title, price, description, and stock state.
FR-05	Cart Management	System shall permit customers to add, modify quantity of, and remove items from an active cart.
FR-06	Cart Validation	System shall reject item quantity additions that exceed current available inventory (`StockQty`).
FR-07	Checkout Processing	System shall capture recipient name, street address, city, and postal code during checkout.
FR-08	Payment Gateway	System shall transmit payment parameters to external Payment Service and await confirmation.
FR-09	Order Generation	System shall generate a unique `OrderID` and save order items upon payment authorization.
FR-10	Inventory Update	System shall automatically decrement product `StockQty` by purchased amounts post-payment.
FR-11	Order Tracking	System shall allow customers to view order history logs and real-time delivery status updates.
FR-12	Admin Catalog	System shall allow Administrators to perform full CRUD operations on product items and categories.
FR-13	Admin Inventory	System shall permit Administrators to manually adjust product inventory stock quantities.
FR-14	Admin Fulfillment	System shall allow Administrators to transition order states (`Pending`, `Shipped`, `Delivered`, `Cancelled`).

2.5 NON-FUNCTIONAL REQUIREMENTS (NFR)

Usability & Interface: Responsive web layout enabling checkout completion in 3 or fewer navigation clicks from cart view.

Performance & Latency: Product searches and catalog loads execute under 2.0 seconds at peak concurrent load (500 users).

Security & Privacy: Mandatory bcrypt password hashing (work factor 12) and HTTPS TLS 1.3 encryption across all network routes.

Reliability & Integrity: Transactional atomic rollback—failed payments enforce zero order creation and zero stock decrements.

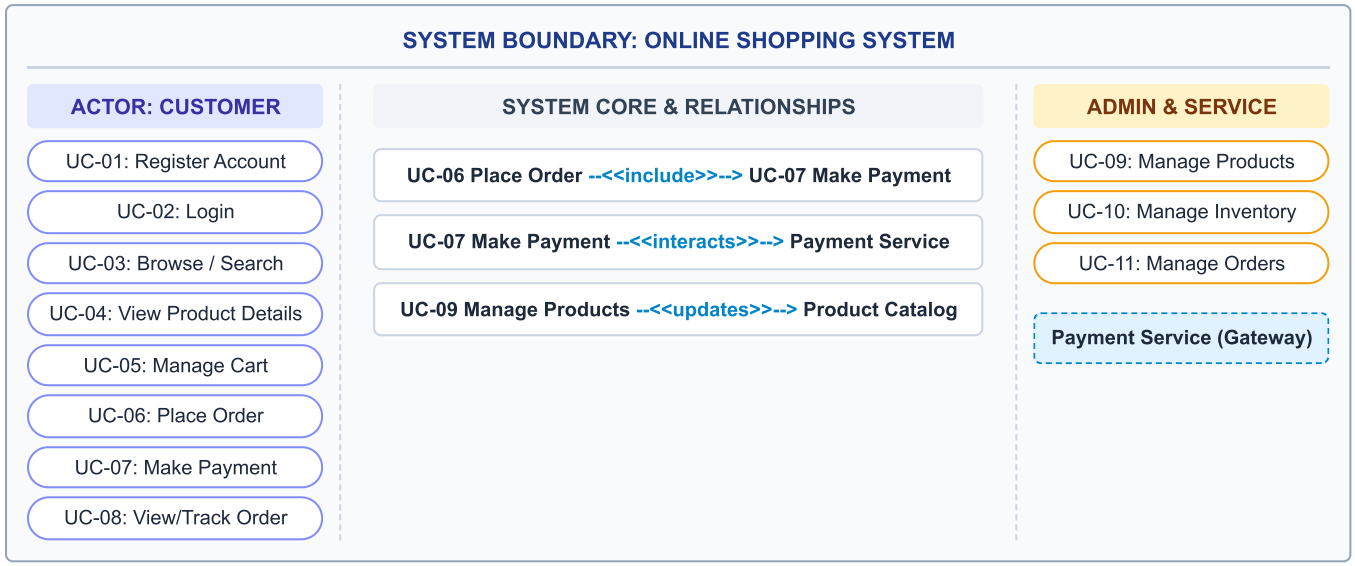
4. USE CASE INVENTORY OVERVIEW

To capture all operational functional requirements, 11 primary use cases have been defined across Customer and Administrator actors within the system boundaries.

UC ID	Use Case Name	Primary Actor	Brief Objective Description
UC-01	Register Account	Customer	Creates a persistent user profile with encrypted credentials.
UC-02	Login	Customer / Admin	Authenticates credentials to open a stateful session.
UC-03	Browse/Search Products	Customer	Queries active catalog items by keywords and categories.

5. UML USE CASE DIAGRAM

Figure 5.1: UML Use Case Diagram Architecture



6. DETAILED USE CASE SPECIFICATIONS (UC-01 TO UC-04)

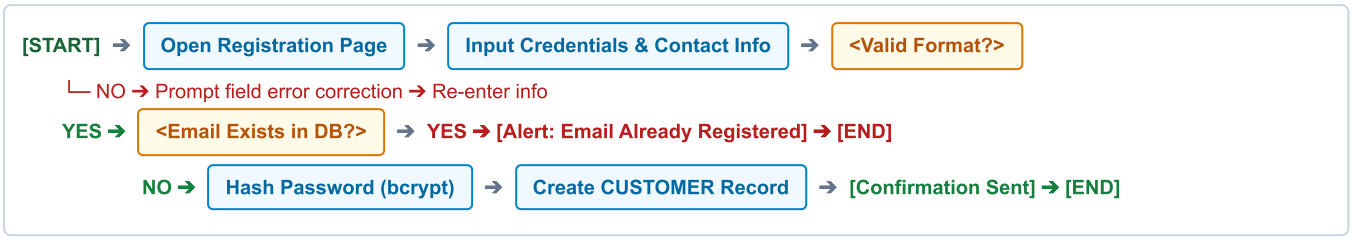
UC ID	Use Case Name	Actors	Flow Specifications & Exception Rules
UC-01	Register Account	Primary: Customer	**Pre:** Visitor on registration page. **Main:** 1. Enters Name, Email, Password, Phone. 2. System validates format and email uniqueness. 3. Hashes password via bcrypt. 4. Saves `CUSTOMER` row. **Except:** Duplicate email triggers error prompt; invalid format forces field correction.
UC-02	Login	Primary: Customer, Administrator	**Pre:** User possesses registered profile. **Main:** 1. Enters Email and Password. 2. System fetches user record and compares password hash. 3. Issues authenticated session token. **Except:** Credential mismatch displays generic "Invalid email or password" alert.
UC-03	Browse/Search Products	Primary: Customer	**Pre:** Products exist in catalog. **Main:** 1. User enters keywords or selects Category. 2. System queries active products (`IsActive=1`). 3. Renders product grid. **Except:** No match returns empty result notice with suggestion to broaden query terms.
UC-04	View Product Details	Primary: Customer	**Pre:** Product list visible on screen. **Main:** 1. User clicks product title/image. 2. System fetches `PRODUCT` row by ID. 3. Renders full description, price, SKU, and available `StockQty`. **Except:** Deactivated product redirects to catalog with unavailable notification.

6. DETAILED USE CASE SPECIFICATIONS (UC-05 TO UC-11)

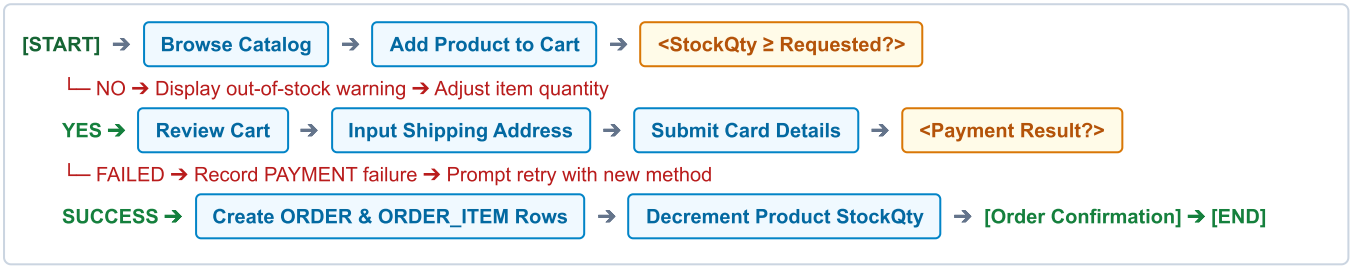
UC ID	Use Case Name	Actors	Flow Specifications & Exception Rules
UC-05	Manage Cart	Primary: Customer	**Pre:** User browsing product pages. **Main:** 1. Selects item quantity and adds to cart. 2. System verifies requested quantity ≤ `StockQty`. 3. Updates session cart subtotal. **Except:** Quantity > stock rejects change with "Insufficient stock available" message.
UC-06	Place Order	Primary: Customer	**Pre:** Cart contains items; user authenticated. **Main:** 1. User reviews cart and inputs shipping address. 2. Triggers UC-07. 3. Upon payment authorization, generates `ORDER` and `ORDER_ITEM` rows. 4. Decrements stock via UC-10. **Except:** Payment rejection aborts order creation; cart state remains unchanged.
UC-07	Make Payment	Primary: Customer Support: Payment Service	**Pre:** Total order amount calculated. **Main:** 1. Customer enters card details. 2. System forwards encrypted payload to Payment Service. 3. Service returns `SUCCESS` token. 4. Logs `PAYMENT` record. **Except:** Gateway failure records status `Failed` and prompts user for retry.
UC-08	View/Track Order	Primary: Customer	**Pre:** Customer has placed past orders. **Main:** 1. Accesses order profile dashboard. 2. System queries orders matching `CustomerID`. 3. Displays items, costs, and shipping tracking state. **Except:** Zero prior orders renders empty historical log notice.
UC-09	Manage Products	Primary: Admin	**Pre:** Admin authenticated. **Main:** 1. Admin inputs SKU, Name, Description, Price, CategoryID. 2. System validates SKU uniqueness. 3. Inserts/updates `PRODUCT` database row. **Except:** Duplicate SKU returns validation error "SKU already exists".
UC-10	Manage Inventory	Primary: Admin	**Pre:** Target product exists. **Main:** 1. Admin updates `StockQty` value. 2. System verifies integer value ≥ 0. 3. Overwrites column value in database catalog. **Except:** Negative input values trigger validation restriction.
UC-11	Manage Orders	Primary: Admin	**Pre:** Orders exist in system. **Main:** 1. Admin reviews order queue. 2. Modifies status field (e.g., `Pending` → `Shipped`). 3. System persists update. **Except:** Invalid lifecycle transitions (e.g., `Cancelled` → `Shipped`) blocked.

7. BUSINESS PROCESS MODELS (BPM)

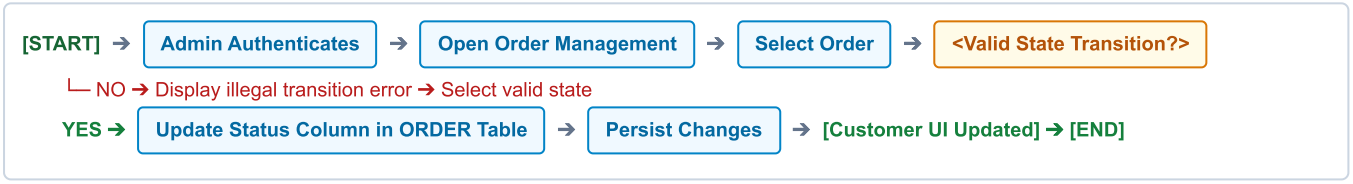
BPM-01: Customer Registration Workflow



BPM-02: Product Purchasing & Payment Workflow



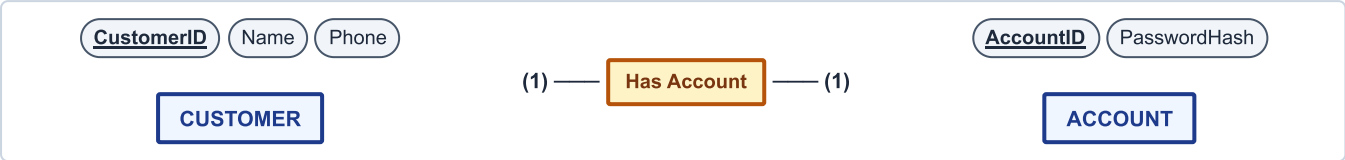
BPM-03: Administrator Order Fulfillment Workflow



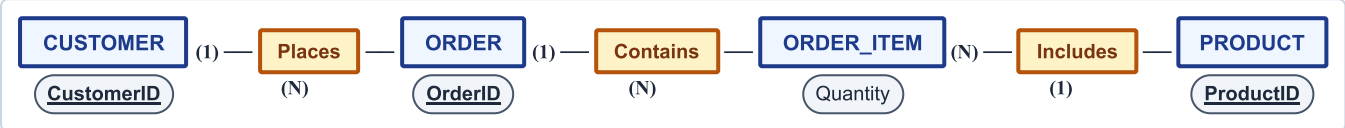
8. CHEN ENTITY-RELATIONSHIP DIAGRAMS (CHEN ERD)

Chen notation models conceptual data structures using rectangles (entities), ovals (attributes, underlined primary keys), diamonds (relationships), and explicit cardinalities.

Chen ERD #1: Customer Account Authentication Mapping



Chen ERD #2: Purchasing Workflow Core Entities



Chen ERD #3: Catalog Structural Classification



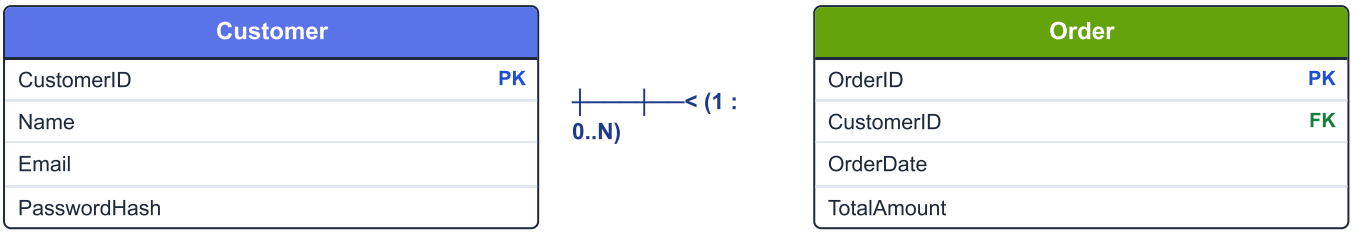
Chen ERD #4: Transaction Fulfillment & Logistics



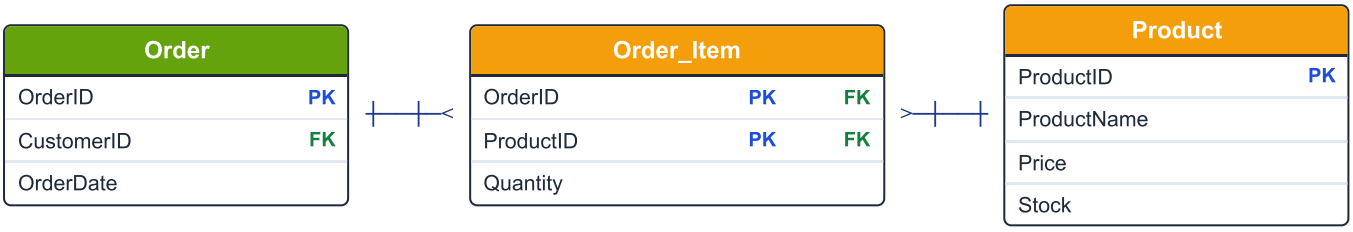
9. INFORMATION ENGINEERING ERDS (IE ERD)

We created our IE ERDs using color-coded entity tables (blue for Customer, green for Order, yellow for Product & Items) with primary key (PK) and foreign key (FK) indicator columns on the right side, connected with crow's foot notation.

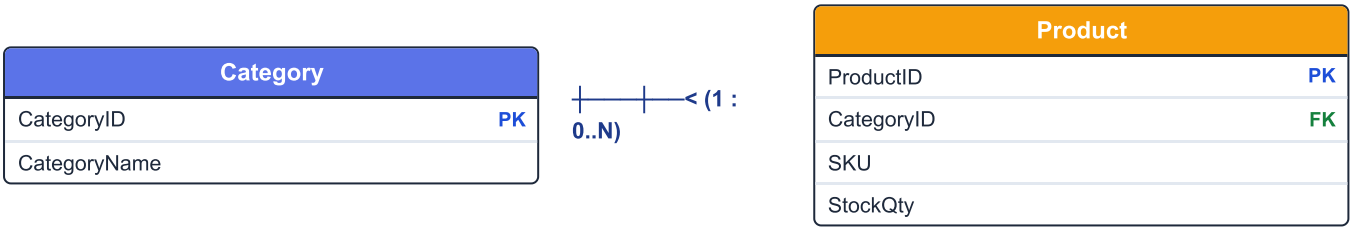
IE ERD #1: Customer to Order Relationship



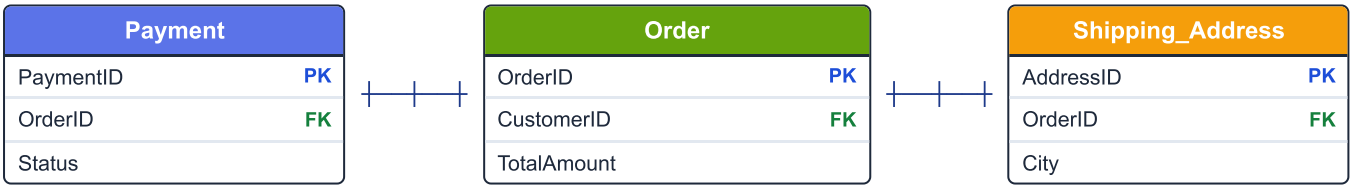
IE ERD #2: Order, Order_Item, and Product Linkage



IE ERD #3: Category to Product Mapping



IE ERD #4: Order, Payment, and Shipping Address (1 : 1)



10. COMPLETE DATABASE DATA DICTIONARY

Entity	Attribute	Data Type	Key	Null	Field Specification Description
CUSTOMER	CustomerID	INT AUTO	PK	NO	Unique primary key for customer profile.
	Name	VARCHAR(100)	-	NO	Full legal name of registered user.
	Email	VARCHAR(150)	-	NO	Unique email address for authentication.
	PasswordHash	VARCHAR(255)	-	NO	Bcrypt cryptographic password hash.
	Phone	VARCHAR(20)	-	YES	Primary contact phone number.
CATEGORY	CategoryID	INT AUTO	PK	NO	Unique identifier for catalog category.
	CategoryName	VARCHAR(100)	-	NO	Category designation (e.g., Electronics).
PRODUCT	ProductID	INT AUTO	PK	NO	Unique identifier for catalog product.
	CategoryID	INT	FK	NO	Refers to parent `CATEGORY(CategoryID)`.
	SKU	VARCHAR(50)	-	NO	Unique Stock Keeping Unit inventory code.
	ProductName	VARCHAR(150)	-	NO	Display name of catalog item.
	Description	TEXT	-	YES	Full product specifications and details.
	Price	DECIMAL(10,2)	-	NO	Current retail unit selling price (> 0.00).
	StockQty	INT	-	NO	Available units on hand (≥ 0).
ORDER	OrderID	INT AUTO	PK	NO	Unique global purchase order ID.
	CustomerID	INT	FK	NO	Refers to purchasing `CUSTOMER(CustomerID)`.
	OrderDate	DATETIME	-	NO	Timestamp when checkout completed.
	TotalAmount	DECIMAL(10,2)	-	NO	Final monetary total of transaction.
	Status	VARCHAR(30)	-	NO	State: Pending, Shipped, Delivered, etc.
ORDER_ITEM	OrderItemID	INT AUTO	PK	NO	Unique identifier for cart line item.
	OrderID	INT	FK	NO	Refers to parent `ORDER(OrderID)`.
	ProductID	INT	FK	NO	Refers to purchased `PRODUCT(ProductID)`.
	Quantity	INT	-	NO	Number of units ordered in line item.
	UnitPrice	DECIMAL(10,2)	-	NO	Historical price per unit at purchase time.

Entity	Attribute	Data Type	Key	Null	Field Specification Description
PAYMENT	PaymentID	INT AUTO	PK	NO	Unique financial payment transaction ID.
	OrderID	INT	FK	NO	Refers to associated `ORDER(OrderID)`.
	PaymentDate	DATETIME	-	NO	Timestamp of gateway authorization response.
	Amount	DECIMAL(10,2)	-	NO	Monetary total authorized and charged.
	Status	VARCHAR(30)	-	NO	State: Completed, Failed, Refunded.
SHIPPING_ADDRESS	AddressID	INT AUTO	PK	NO	Unique destination address record ID.
	OrderID	INT	FK	NO	Refers to target `ORDER(OrderID)`.
	RecipientName	VARCHAR(100)	-	NO	Full name of parcel delivery recipient.
	Street	VARCHAR(255)	-	NO	Street address and apartment/suite details.
	City / PostalCode	VARCHAR(50)	-	NO	Destination city and ZIP/Postal Code.

11. RELATIONAL CARDINALITY ANALYSIS

- **Customer → Order (1:N):** One customer places 0..N orders; each order is tied strictly to 1 customer.
- **Category → Product (1:N):** One category classifies 0..N products; each product belongs to exactly 1 category.
- **Order → Order_Item (1:N):** One order contains 1..N order items; each item line belongs to 1 parent order.
- **Product → Order_Item (1:N):** One product is referenced in 0..N order items; each item line references 1 product.
- **Order → Payment (1:1):** One order maps strictly to 1 payment record; each payment references 1 order.
- **Order → Shipping Address (1:1):** One order links to 1 destination address; each address maps to 1 order.

12. CORE SYSTEM BUSINESS RULES (BR)

BR-01 (Email Uniqueness): Customer email addresses must be strictly unique within database records.

BR-02 (Authentication Guard): Customers must be logged in before proceeding through checkout.

BR-03 (Category Association): Products must belong to an active category before catalog display.

BR-04 (Price Enforcement): Product unit price must always be strictly greater than zero (`Price > 0`).

BR-05 (Stock Quantity Limit): Cart requested quantity cannot exceed current available `StockQty`.

BR-06 (Order Item Floor): An order must contain at least one valid line item to complete creation.

BR-07 (Payment Requirement): Order status cannot transition to Shipped without `Completed` payment status.

BR-08 (Inventory Isolation): Stock replenishment updates are restricted exclusively to Administrator roles.

BR-09 (Data Isolation): Customers may only query order records matching their authenticated `CustomerID`.

BR-10 (Transaction Rollback): Payment failures abort order creation and prevent stock adjustments.

BR-11 (SKU Uniqueness): Catalog products must possess globally unique SKU codes.

BR-12 (Price Snapshot): Order item stores price snapshot at purchase time, ignoring future catalog updates.

14. END-TO-END SYSTEM INTEGRATION WORKFLOW

```
[START] --> (Visitor Accesses Platform) --> (UC-01: Register / UC-02: Authenticate) | v (UC-03: Search / Browse (
 --NO--> [Display Stock Deficit Alert] --> (Adjust Quantity) | YES v (Proceed to Checkout & Enter Shipping Addres

  --FAILED--> [Record Failed PAYMENT] --> (Prompt Retry) | SUCCESS v (UC-06: Create ORDER & ORDER_ITEM Records ir
```

13. ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

System Role	Authorized System Capabilities and Endpoint Privileges
Visitor	Browse public catalog, search items by keyword, apply category filters, view product details, register new account.
Customer	All Visitor privileges + authenticate, manage session cart, enter shipping details, process payment, view own order history, track order status.
Administrator	Authenticate admin portal, execute CRUD operations on products/categories, adjust inventory stock levels manually, review system-wide orders, execute status transitions.

16. CONCLUSION & TEAM VERIFICATION CHECKLIST

In this Capstone 1 report, our team (Galautdinova Ruzana, Rugacheva Arina, and Kan Daniil) completed a system analysis and design for an Online Shopping System. We analyzed all system requirements, modeled the core purchasing workflows using 11 Use Case specifications, mapped out 3 Business Process Models (BPM), and designed 4 Chen ERDs alongside 4 Information Engineering (IE) ERDs. This report serves as our complete foundation for subsequent database implementation and software development.

Deliverable Requirement Metric	Required Benchmark	Our Project Count	Compliance
Use Case Specifications & Modeling	> 3 Use Cases	11 Use Cases	**SATISFIED**
Business Process Models (BPM)	> 2 Workflows	3 BPM Workflows	**SATISFIED**
Chen Notation ERDs	> 3 Diagrams	4 Chen Diagrams	**SATISFIED**
IE Crow's Foot ERDs	> 3 Diagrams	4 IE Diagrams	**SATISFIED**
Database Dictionary & Rules	Complete Schema	7 Tables / 12 Rules	**SATISFIED**